

MECHANIC SETTLEMENT, NB, Canada

WMO: 717870

Lat: 45.6936N

Lon: 65.1650W

Elev: 403

StdP: 96.58

Time Zone: -4.00 (NAA)

Period: 06-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -22.0	(c) -19.6	(d) -26.5	(e) 0.4	(f) -21.1	(g) -23.7	(h) 0.5	(i) -18.1	(j) 12.1	(k) -4.3	(l) 11.3	(m) -5.7	(n) 5.9	(o) 290	(p) 0.668

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 8.1	(c) 26.0	(d) 19.2	(e) 24.5	(f) 18.4	(g) 23.1	(h) 17.4	(i) 20.6	(j) 23.9	(k) 19.8	(l) 22.6	(m) 19.0	(n) 21.5	(o) 4.4	(p) 240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.7	15.1	21.5	18.9	14.4	20.9	18.0	13.6	20.2	61.4	24.0	58.6	22.7	55.7	21.4	23.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 10.4	(b) 9.3	(c) 8.4	(e) -25.5	(f) 29.1	(g) 2.4	(h) 1.4	(i) -27.2	(j) 30.2	(k) -28.7	(l) 31.0	(m) -30.0	(n) 31.8	(o) -31.8	(p) 32.8		
(a) 10.4	(b) 9.3	(c) 8.4	(e) -25.6	(f) 22.2	(g) 2.4	(h) 0.6	(i) -27.3	(j) 22.6	(k) -28.7	(l) 22.9	(m) -30.0	(n) 23.3	(o) -31.7	(p) 23.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	5.0	-8.5	-7.7	-3.8	2.9	9.3	13.9	18.3	17.6	13.6	7.4	1.3	-5.0
	DBStd	10.42	6.46	5.64	5.14	4.30	4.05	3.38	2.73	2.67	3.82	4.29	5.16	5.92
	HDD10.0	2626	573	497	428	216	63	7	0	0	11	103	264	464
	HDD18.3	4930	831	730	686	462	281	137	34	46	149	340	512	723
	CDD10.0	804	0	0	1	4	41	124	257	235	117	22	2	0
	CDD18.3	64	0	0	0	0	1	4	32	22	6	0	0	0
	CDH23.3	273	0	0	0	1	11	27	131	78	26	1	0	0
	CDH26.7	26	0	0	0	0	1	1	14	5	4	0	0	0
Wind	WSAvg	4.4	5.1	5.0	5.0	4.6	4.2	3.9	3.8	3.7	4.0	4.5	4.8	4.8
Precipitation	PrecAvg	1316	128	86	121	102	111	93	91	76	117	131	124	126
	PrecMax	1593	183	161	239	166	228	240	176	157	277	254	207	256
	PrecMin	980	66	28	20	51	45	38	40	16	44	27	65	46
	PrecStd	148	40	33	48	29	47	49	35	40	59	63	42	44
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.1	8.5	15.3	18.5	24.7	26.3	28.6	27.4	26.2	20.1	15.3	11.2
		MCWB	9.9	8.1	11.2	11.2	15.8	17.7	19.9	20.5	20.3	16.1	14.9	11.1
	2%	DB	6.5	4.9	7.2	14.3	21.2	23.4	26.2	25.4	23.2	17.4	13.3	8.1
		MCWB	6.3	4.2	5.0	9.8	13.7	16.5	19.5	19.2	18.4	14.8	12.5	7.6
	5%	DB	3.3	1.9	5.2	12.0	18.2	21.3	24.7	23.9	21.0	15.5	11.3	5.4
		MCWB	2.9	0.8	3.2	9.0	12.3	15.5	18.6	18.1	17.2	13.8	10.0	5.1
	10%	DB	0.4	0.1	3.1	10.0	16.0	19.7	23.3	22.4	19.1	13.8	9.1	3.1
		MCWB	0.0	-0.5	1.1	7.0	11.0	14.2	17.9	17.3	16.5	12.0	7.9	2.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.1	8.3	11.2	14.0	17.7	19.8	21.8	21.6	20.8	17.3	15.2	11.1
		MCDB	9.3	8.3	15.2	14.5	22.6	23.5	25.2	25.2	25.3	18.1	15.3	11.2
	2%	WB	6.0	4.4	6.2	11.7	15.8	18.2	20.7	20.5	19.4	15.8	13.1	7.9
		MCDB	6.2	4.7	6.9	13.1	18.7	20.6	24.1	23.3	21.6	16.4	13.2	8.0
	5%	WB	2.5	1.2	3.7	9.8	14.1	17.0	20.0	19.6	18.4	14.1	10.6	5.2
		MCDB	2.8	1.7	4.4	11.1	16.6	19.6	22.9	22.2	19.9	15.1	10.9	5.3
	10%	WB	0.0	-0.4	1.3	7.4	12.4	15.8	19.2	18.6	17.3	12.3	8.0	2.8
		MCDB	0.3	-0.1	2.7	9.3	15.2	18.3	21.9	21.1	18.7	13.4	8.8	3.1
Mean Daily Temperature Range	5% DB	MDBR	8.3	8.1	8.1	8.1	9.4	8.3	8.1	8.0	8.1	7.2	6.8	7.4
		MCDBR	11.5	9.5	10.0	11.1	12.2	10.8	9.9	9.6	9.5	8.4	9.1	9.9
		MCWBR	11.3	8.5	8.1	7.9	7.1	6.0	5.2	5.3	5.8	6.5	8.5	10.1
	5% WB	MCDBR	11.0	9.5	9.7	10.0	10.6	9.2	8.6	8.3	8.1	8.1	8.9	9.8
		MCWBR	11.3	9.2	9.3	8.6	7.5	6.0	5.2	5.3	6.1	7.3	9.2	10.1
Clear-Sky Solar Irradiance	taub	0.274	0.274	0.305	0.331	0.353	0.378	0.405	0.375	0.342	0.313	0.295	0.275	
	taud	2.437	2.437	2.443	2.431	2.408	2.356	2.312	2.398	2.475	2.558	2.537	2.494	
	Ebn at Noon	845	919	931	930	914	888	857	874	877	863	817	808	
	Edh at Noon	69	83	96	107	114	120	124	110	93	74	63	59	
All-Sky Solar Radiation	RadAvg	1.45	2.27	3.44	4.29	4.99	5.39	5.54	4.93	3.93	2.50	1.40	1.08	
	RadStd	0.13	0.18	0.22	0.40	0.52	0.45	0.37	0.34	0.30	0.26	0.15	0.08	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(a) Station Only	(b) Regional (45 neighbors)	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A
(a) Station Only	(b) Regional (45 neighbors)	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) +0.40	(j) N/A	(k) N/A	(l) +48	(m) N/A

Nomenclature: See separate page